

Seasonal and comparative evidence of adaptive gene expression in mammalian brain size plasticity

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eLife Assessment

This study presents **valuable** findings related to seasonal brain size plasticity in the Eurasian common shrew (*Sorex araneus*), which is an excellent model system for these studies. The evidence supporting the authors' claims is **convincing**. The work will be of interest to biologists working on neuroscience, plasticity, and evolution.

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Abstract

Contrasting almost all other mammalian wintering strategies, Eurasian common shrews, *Sorex araneus*, endure winter by shrinking their brain, skull, and most organs, only to then regrow to breeding size the following spring. How such tiny mammals achieve this unique brain size plasticity while maintaining activity through the winter remains unknown. To discover potential adaptations underlying this trait, we analyzed seasonal differential expression in the shrew hypothalamus, a brain region that both regulates metabolic homeostasis and drastically changes size, and compared hypothalamus expression across species. We discovered seasonal variation in suites of genes involved in energy homeostasis and apoptosis, shrew-specific upregulation of genes involved in the development of the hypothalamic blood brain barrier and calcium signaling, as well as overlapping seasonal and comparative gene expression divergence in genes implicated in the development and progression of human neurological and metabolic disorders, including *CCDC22*. With high metabolic rates and facing harsh winter conditions, *Sorex araneus* have evolved both adaptive and plastic mechanisms to sense and regulate their energy budget. Many of these expression changes mirrored those identified in human neurological and metabolic disease, highlighting the interactions between metabolic homeostasis, brain size plasticity, and longevity.

Introduction

Typical mammalian brain development consists of unidirectional postnatal growth until reaching an adulthood maximum (1, 2), but the Eurasian common shrew, *Sorex araneus*, seasonally changes the size of its body, most organs, skull, and —especially— its brain (3–6). In the most acute case of Dehnel's phenomenon, or seasonal size plasticity, the common shrew first grows to an initial summer maximum as a juvenile, then shrinks in autumn losing up to 18% body mass, ~20% skull volume, and 26% brain mass, reaching its winter size minimum (3–6). Then, in the spring prior to mating, these shrews partially or fully regrow these organs, reaching a second maximum. Dehnel's phenomenon is hypothesized to be an adaptation to decrease energy demand without slowing metabolism through plasticity, improving shrew fitness during the low temperatures and limited food supply of winter (3, 4, 6–12). Corroborating this adaptive plasticity hypothesis, Dehnel's phenomenon anticipates winter conditions and is both geographically variable (4) and modulated by environmental temperature (3). Although we have begun to reveal the molecular mechanisms of Dehnel's phenomenon (12), the comparative context of proposed adaptations, and commonality of processes across brain regions remain unknown. Elucidating both the genetic basis and evolutionary mechanisms underlying Dehnel's phenomenon will illuminate the molecular adaptations involved in this unusual case of mammalian brain plasticity.

As a hypothesized wintering strategy, energy metabolism is key to the evolution of Dehnel's phenomenon in *S. araneus* (11). Tiny mammals have limited options to survive winter conditions (13), and Dehnel's phenomenon is exceedingly rare as a wintering strategy. While many mammals seasonally migrate away from low-temperature, low-productivity environments, some endure these conditions by reducing their energy requirements through hibernation; a phenotype that has independently evolved many times (14–16). The evolution and regulation of hibernation relies upon a suite of metabolic genes (17, 18), including those associated with thermogenesis, circadian rhythms, and feeding behavior. Compared to most species, *S. araneus* has astronomically large energy demands, with one the highest basal metabolic rates per unit of body mass identified in any mammal (19, 20). To meet these energy requirements, *Sorex araneus* forage and feed constantly. Together with brief lifespans (~1 year) that nonetheless require surviving a single winter (21, 22), these extreme metabolic demands constrain survival options for *S. araneus*. Thus, by decreasing resources devoted to energetically expensive tissue such as the brain (23), Dehnel's phenomenon bypasses these constraints and expands the niche of a typical small mammal during winter while allowing this fast-living, predatory shrew to remain active year round.

We hypothesize Dehnel's phenomenon, which involves metabolic shifts and inter-organ communication (12), has evolved through the combination of both physiological plasticity and genetic adaptation. This idea is twofold; phenotypic plasticity is not only adaptive in itself as the evolved spectrum of a trait, but can also allow populations to persist in varying environments, leading to ancillary evolutionary adaptation when given enough time and strength of selection (24). Thus, while temporal analysis of shrew expression can elucidate putatively adaptive plastic regulation, it will miss adaptive canalization of gene expression that has contributed to the evolution of Dehnel's phenomenon. Therefore, plastic regulatory mechanisms and proposed adaptations can be further tested via cross-species comparisons of expression divergence. These cross-species comparisons treat expression as a trait and have been used to characterize diverse vertebrate adaptations including thermal tolerance and vision loss in fish (25–27), the effects of whole genome duplication on expression (28–32), sex specific alternative splicing in birds (33), and patterns of expression evolution in mammalian organs (34–37). By incorporating these comparative approaches with seasonally varying gene expression, we can further elucidate the role of plasticity and adaption in both the regulation and evolution of Dehnel's phenomenon.

The hypothalamus is an intriguing candidate brain region for comparative analysis, as it is both the brain region with the greatest seasonal size change in shrews, and the center of bodily homeostatic maintenance across mammals. In shrews, the hypothalamus undergoes a 31.6% volume reduction in autumn through winter, followed by a 47.8% increase in spring (38). Across all mammals, the hypothalamus plays a pivotal role in the maintenance of energy budgets, with functions influencing: 1) energy intake and feeding behavior, 2) energy expenditure and metabolic rate, and 3) energy deficits and storage. While such functions are important for all mammals, these are critical while wintering to maintain stable internal body temperatures as external temperatures decrease. For example, the hypothalamus can activate the sympathetic nervous system as temperatures decrease, stimulating winter thermogenesis. Thus, shrews may deploy typical mammalian hypothalamic plasticity concurrent with seasonal size plasticity. Alternatively, and considering how rare Dehnel's phenomenon is, shrews may have evolved divergent approaches, akin to naked mole rat traits, a species that lives under consistent hypoxic conditions resulting in adaptations associated with decreased oxygen and blood flow across the brain (39). Comparative approaches, then, may identify novel adaptations that contribute to the evolution of Dehnel's phenomenon in the shrew hypothalamus.

As the brain region able to quickly respond to change, the hypothalamus has remarkable plastic capabilities associated with metabolism (40), especially as environmental conditions vary between seasons and can be unpredictable. While the central nervous system (CNS) acts somewhat independently of bodily metabolism through the protective effects of the blood brain barrier (BBB), the CNS is still sensitive to bodily metabolic change. We hypothesize hypothalamic plasticity is central to the evolution and regulation of size plasticity through both environmental sensing mechanisms and signaling responses to these stimuli. Specifically, we hypothesize adaptations in the shrew: 1) hypothalamic BBB associated with dynamic metabolic fluctuations, 2) sensing of metabolic state and seasonal size through hormonal signals, mediated by BBB-crossing of insulin, ghrelin, and leptin, and 3) responses to metabolic fluctuations involving ion-dependent signaling. With limited energy inertia, shrews must continuously sense their peripheral metabolism, and a combination of receptors and selective BBB permeability in the hypothalamus allows certain molecules to relay information from the peripherals to the brain. Among the best characterized of these are the peripheral hormones insulin, ghrelin, and leptin, the latter known to excite anorexigenic and inhibit orexigenic neurons in the arcuate nucleus of the hypothalamus. Finally, according to Ramon y Cajal's neuronal doctrine (41), differences in brain functional responses may stem from altered synaptic firing. Thus, inter- and intracellular ion concentrations play a large role in the communication between neuronal networks by propagating signals in response to perceived environmental stimuli.

To test these hypotheses, we analyzed both the seasonal and phylogenetic variation in shrew hypothalamic expression to detect signals of adaptive plasticity. First, we aimed to identify differential expression of genes across five seasons of Dehnel's phenomenon that might promote regulatory responses to seasonal variation and can be functionally validated with cell line perturbations. Second, this analysis was paired with a comparative transcriptomics approach using hypothalamic expression data from 15 additional mammal species. These analyses infer putative adaptation by testing for branch-specific expression shifts using Ornstein-Uhlenbeck models. The objective of these evolutionary analyses was to quantify lineage-specific hypothalamic expression changes in *S. araneus*, with evolutionary expression divergence consistent with selection. Finally, by comparing individual genes and associated pathways from these two analyses, we can determine potential adaptive plasticity, indicating mechanisms that both potentially regulate size plasticity and were selected for higher expression in the evolution of Dehnel's phenomenon. Our expression results implicate several key processes in Dehnel's phenomenon, including seasonal plasticity in feeding behavior, enhanced modulation of both hypothalamic blood brain barrier and downstream signaling, and plastic apoptosis responses.

Results

Temporal Expression

We identified several known neural signaling pathways correlated with patterns of seasonal hypothalamus expression that may be associated with changes during Dehnel's phenomenon. We began by hierarchically clustering expression to identify patterns of gene expression through time. Most genes did not exhibit much variation through time, indicating constitutive gene expression in the hypothalamus, as only 786 of 19,296 genes passed our filters (>0.5-fold change between any two seasons and 2 samples > 10 normalized reads). We identified 12 distinct clusters of gene expression patterns by bootstrapping (n=20) a gap statistic within cluster distances. Of these 12 clusters, five clusters consisting of 392 genes resembled a large divergence between summer juveniles and the remaining individuals without a second local maximum during size change (Supplemental Figure 2; Clusters 2, 3, 8, 11, 12). These genes likely represent a large developmental shift between recently postnatal shrews and the remainder of individuals compared to shifts associated with seasonality or Dehnel's phenomenon. We then functionally characterized the remaining 394 genes (**Figure 1A** [↗](#)), which exhibited seasonal shifts, with a GO pathway enrichment using the Database for Annotation, Visualization, and Integrated Discovery (DAVID) Gene Functional Classification Tool. Although no enrichment pathway was significant after a Bonferroni correction, 14 pathways were enriched prior to correction ($p < 0.05$; Modified Fisher's Exact Test); with the five pathways with the lowest p-values including fluid shear stress and atherosclerosis, relaxin signaling, neuroactive ligand-receptor interaction, HIF-1 signaling, and PI3K-Akt signaling (**Figure 1B** [↗](#)). Many of these pathways have been implicated in various physiological processes, which suggests that variation identified in the hypothalamus are likely the result of autonomic processes in homeostatic maintenance. Individual enrichments of each pathway can be found in Supplemental Table 2.

We also discovered hundreds of differentially expressed genes between autumn and spring shrews that may mediate phenotypic divergence in both size (shrinkage vs. regrowth) and metabolic changes (liver expression shifts ([12](#) [↗](#))) associated with Dehnel's phenomenon. By comparing the RNA expression of autumn and spring shrews, we found 210 differentially expressed genes (DEGs), with 129 upregulated and 81 downregulated in spring (**Figure 2A** [↗](#)). We ran a GO pathway enrichment to functionally characterize the 210 DEGs in the hypothalamus and found fifteen pathways were enriched ($p < 0.05$; Modified Fisher's Exact Test) with significant genes prior to multiple comparison correction. These pathways were all downregulated in spring and included; pathways in cancer (4.5-fold enrichment, $p < 0.001$), transcriptional misregulation in cancer (8.3-fold enrichment, $p < 0.001$), fluid shear stress and atherosclerosis (9.5-fold enrichment, $p < 0.01$), small cell lung cancer (11.5-fold enrichment, $p < 0.01$), IL-17 signaling (11.3-fold enrichment, $p < 0.01$), cAMP signaling (5.9-fold enrichment, $p < 0.01$), and apoptosis (7.8-fold enrichment, $p < 0.05$) (**Figure 2B** [↗](#)). Pathways in cancer consisted of nine significantly downregulated genes (from DESeq2 Wald test), four of which overlapped with apoptotic pathways including *BCL2L1* (-0.6 Log-Fold Change [LFC] $p_{\text{adj}} < 0.01$), *NFKBIA* (-1.3 LFC, $p_{\text{adj}} < 0.01$), *FOS* (-1.5 LFC, $p_{\text{adj}} < 0.05$), and *GADD45B* (-1.4 LFC, $p_{\text{adj}} < 0.05$) (**Figure 2C** [↗](#)). Differential expression for all genes can be found in Supplemental Data. Overall, the processes identified suggest regulation of cell proliferation (cancer pathways) and death (apoptosis) in the hypothalamus may be associated with phenotypic changes during Dehnel's phenomenon.

Cell Viability Analysis with BCL2L1 Overexpression

We examined the potential functional effects of apoptosis regulating gene, *BCL2L1* (**Figure 2D** [↗](#)), which was differentially expressed between seasons, by propagating an *in vitro* model of domesticated ferret brain cells (MPF-CRL-1656, ATCC). We chose this cell line because there is no

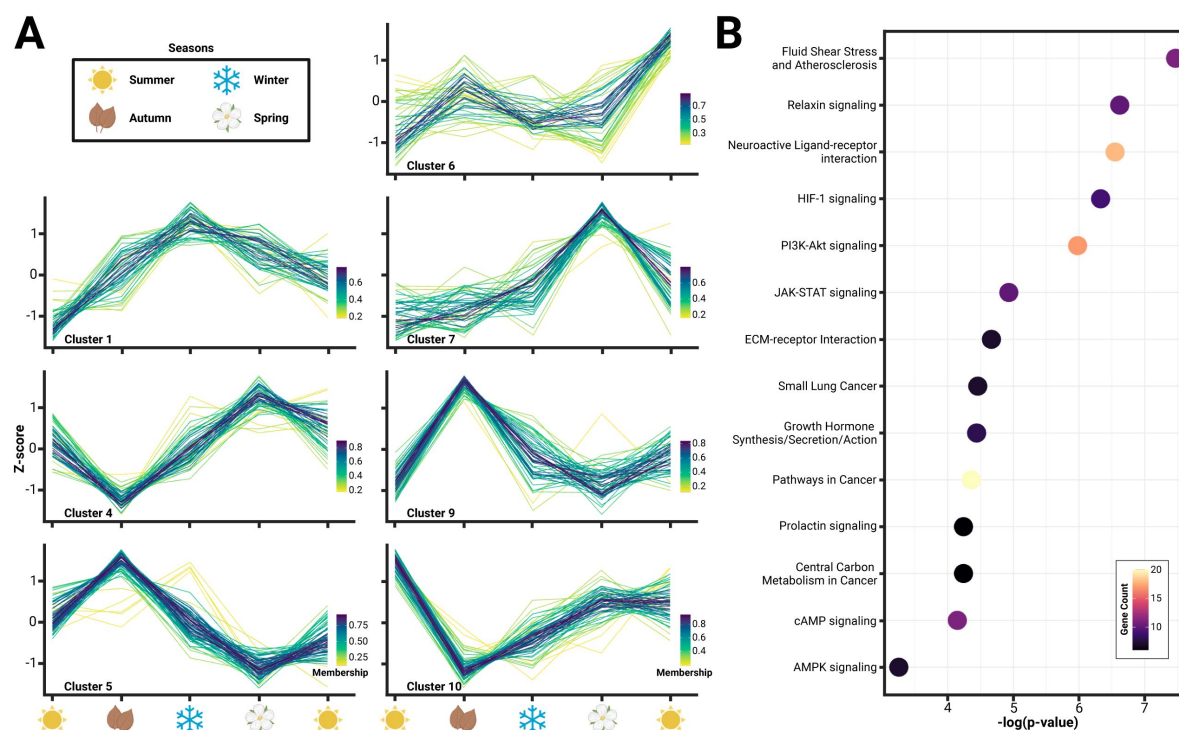


Figure 1

(A) Hierarchical clustering of expression identified 12 distinct clusters, of which seven clusters (1, 4, 5, 6, 7, 9, 10) comprising of 394 genes, showed variation consistent with seasonality or Dehnel's phenomenon. **(B)** Functional characterization of these genes using KEGG GO pathways found an enrichment of 14 pathways ($p < 0.05$), many of which have been implicated in hypothalamic control of homeostatic maintenance, including, relaxin signaling, neuroactive ligand-receptor interaction, HIF-1 signaling, and PI3K-Akt signaling.

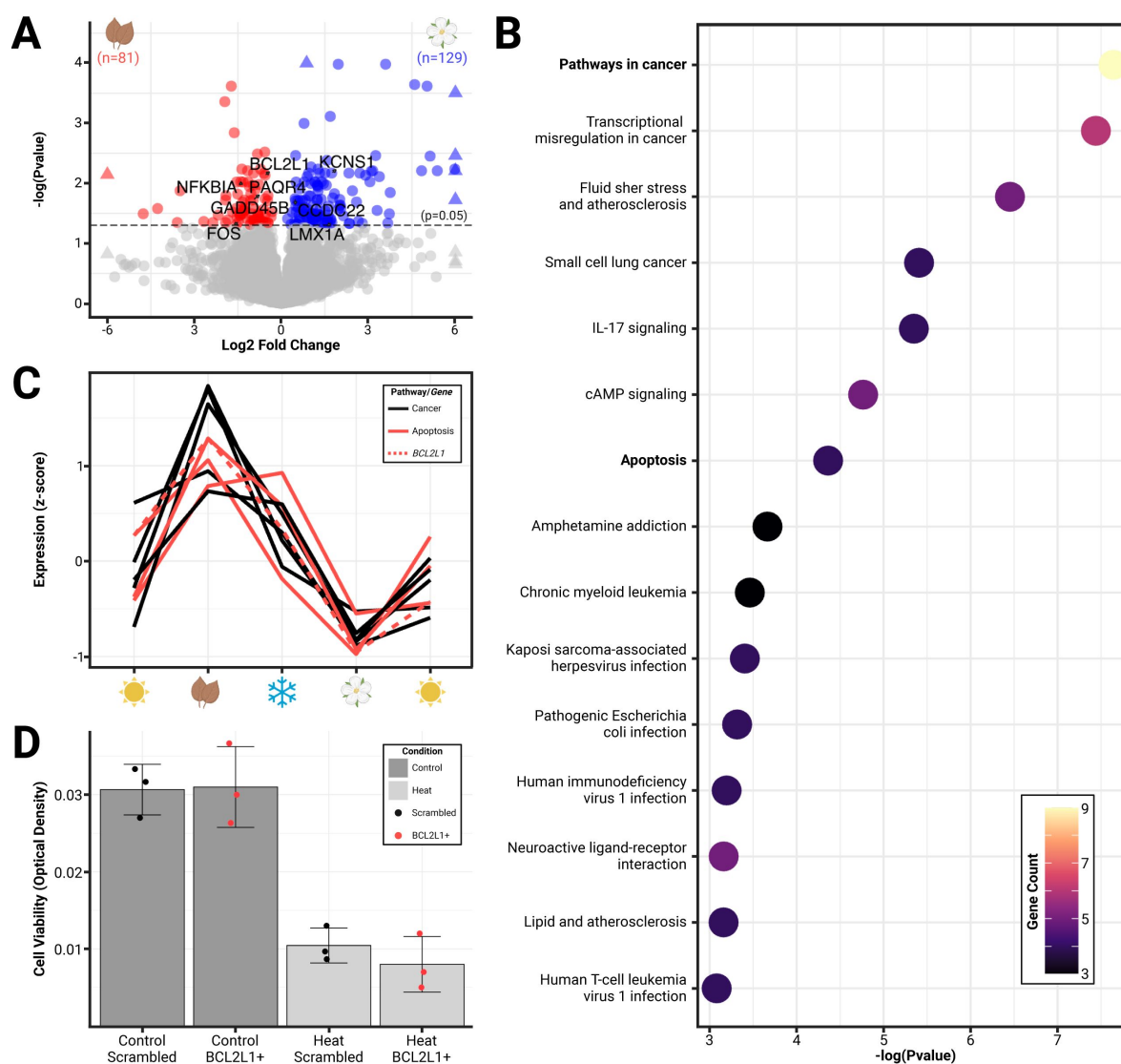


Figure 2

(A) Volcano plot of significant ($p_{\text{adj}} < 0.05$) differentially expressed genes (colored) between phenotypic extremes of hypothalami size change (spring vs autumn) plotted by log fold-change. (B) Pathway enrichment analysis identified 15 pathways to be enriched for differentially expressed genes, including pathways in cancer and apoptosis (downregulated). (C) Patterning of gene expression across seasons of Dehnel's phenomenon for genes found in the cancer (all) and apoptosis (red) pathways, including *BCL2L1* (dashed). (D) Cell viability of *Mustela putorius furo* neural cell lines exposed to four treatments: scrambled *BCL2L1* overexpression, *BCL2L1* overexpression, heat with scrambled *BCL2L1* overexpression, and heat with *BCL2L1* overexpression. Heat significantly reduced the cell viability compared to controls but was not rescued by *BCL2L1* overexpression.

established cell line of *Sorex araneus*. *Mustela putorius furo* also undergoes a Dehnel's-like phenotype (42) and is more closely related to shrews than mice are. MPF cells were transfected with either the anti-apoptotic *BCL2L1* RNA or a scrambled version of *BCL2L1* (s*BCL2L1*) RNA. 24 hours after transfection, cells were exposed to heat stress followed by crystal violet staining. A significant difference in cell viability was observed between the heat scrambled compared to the control scrambled groups ($df=3.56$, $t_{3.56}=8.78$, $p<0.01$), however, there was no significant difference between cells transfected with *BCL2L1* compared to those transfected with s*BCL2L1* when exposed to heat ($df=3.37$, $t_{3.37}=-0.99$, $p=0.39$).

Evolutionary Divergence in Expression

Analyses quantifying branch-specific shifts in expression identified hundreds of genes that were upregulated in the shrew hypothalamus (post pubescent spring individuals) compared to other mammals, suggesting adaptation for increased expression in these genes (Figure 3). EVE models found 222 genes significantly rejected ($p_{adj} < 0.05$) a single expression optimum for all species in favor of a second expression optimum for the shrew lineage. Notably, all second optima (in the shrew lineage) were higher than the first optimum, consistent with results from the original methods manuscript (34). While nonsignificant genes had a mean 0.93-fold change, genes experiencing an expression branch-shift in shrews had a mean 9.66 oTPM-fold change compared to other species. To validate genes with expression branch-shifts as shrew-specific, we ran a dropout test for the shrew lineage, testing for phylogeny-wide evidence of selection for each gene. Of the 6,496 genes tested, 81 genes had significantly ($p_{adj} < 0.05$) lower β (within population variance to between species variance), suggesting selection elsewhere in the phylogeny. None of these genes overlapped with those identified to be differentially expressed in the shrew, further validating the specificity of expression shifts.

We identified several changes associated with processes that may underlie seasonal size plasticity. Pathway enrichment analysis identified two significantly enriched pathways: calcium signaling (2.8-fold enrichment, $p<0.05$) and autophagy (9.1-fold enrichment, $p<0.05$) (Figure 3). Upon manual inspection of the list, we also identified several other potential adaptive processes, including blood-brain-barrier formation and function, feeding behavior and leptin responses, metabolism, and neuroprotection. (Figure 3, Table 1).

Many of these putative adaptations mediate responses to environmental cues centered around energy demands. Four of the genes experiencing branch-shift changes in expression were also differentially expressed between autumn and spring individuals, suggesting not only an adaptive shift in the shrew lineage associated with the evolution of Dehnel's phenomenon, but also a direct molecular mechanism for brain changes. The four genes that overlap between both analyses include (Figure 4): *CCDC22*, which plays an important role in endosomal recycling of membrane proteins (43); *LMX1A*, a transcription factor essential for dopaminergic neuron development (44); *KCNS1*, which encodes a subunit of potassium channels associated with neuronal excitability and pain sensitivity (45); and *PAQR4*, which appears to regulate growth and apoptosis through sphingolipid synthesis in human cancers (46). However, identifying four overlapping genes is not significantly greater than expected by chance (Fisher's Exact Test, $p=0.55$).

Discussion

By characterizing both seasonal and between-species differential expression of the hypothalamus, we generated and probed a unique data set, discovering expression shifts associated with extreme brain and body size plasticity in *Sorex araneus*. While the focus on seasonal hypothalamus expression provided insights into both the regulation of seasonal homeostasis and processes underlying brain size change, evolutionary shifts in expression revealed putative adaptations associated with mammalian brain degeneration and regeneration in a natural system. Our

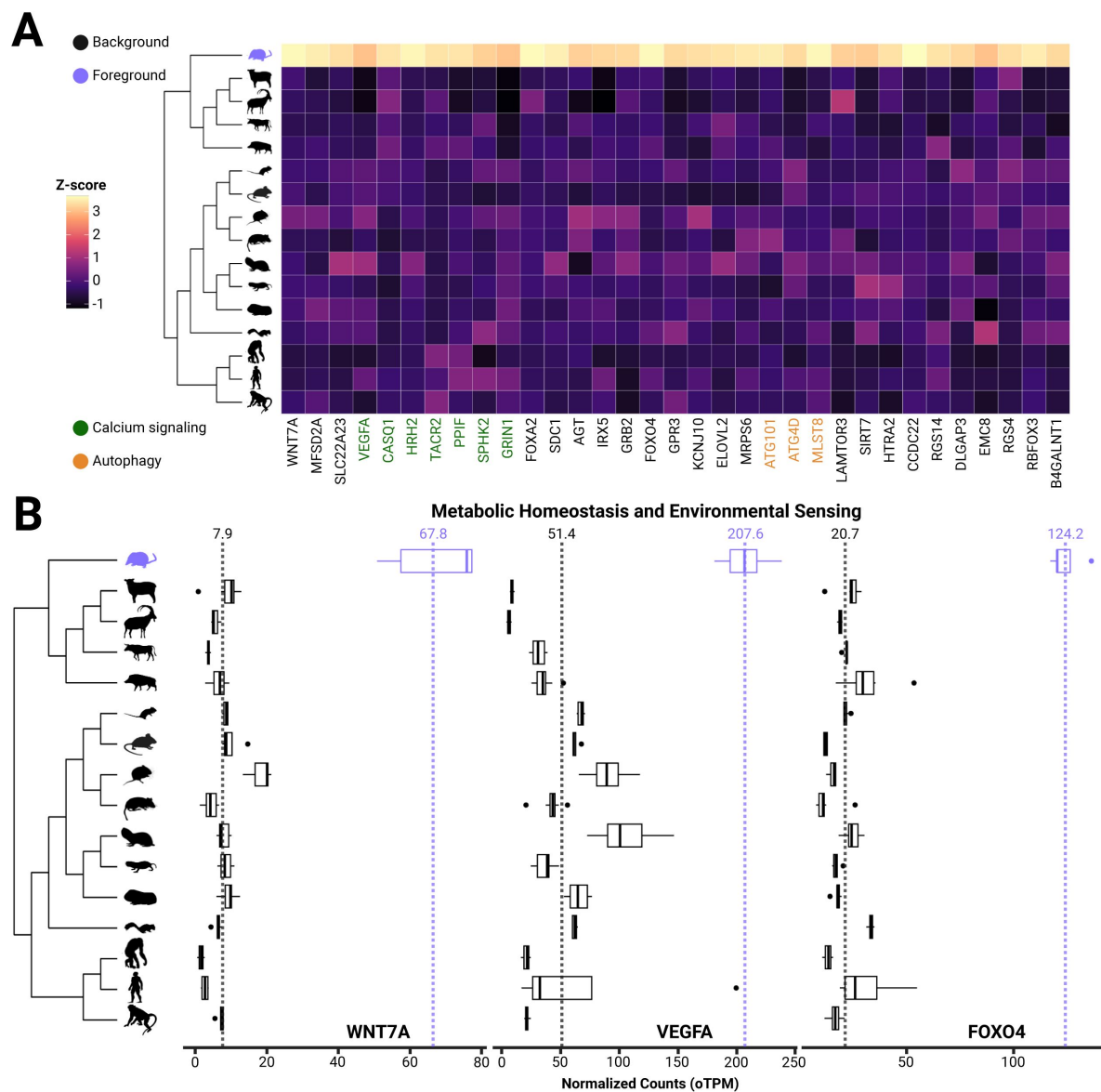


Figure 3

(A) Heatmap and boxplots of genes with shrew-specific upregulation compared to other mammals associated with processes including calcium signaling, autophagy, neurological functions, (B) metabolic homeostasis and environmental sensing.

Gene	Function	FC	P-value	KEGG (p<0.05)
WNT7A	BBB/Neurogenesis/Angiogenesis	8.60	0.000	NA
MFSD2A	Transports DHA across blood brain barrier/Fasting induced	5.22	0.000	NA
SLC22A23	BBB plasticity	4.69	0.000	NA
VEGFA	BBB plasticity/Calcium Signaling	4.04	0.024	Calcium Signaling
CASQ1	Calcium Signaling/myopathy	7.22	0.000	Calcium Signaling
HRH2	Calcium Signaling/Sleep related/Histaminergic neurons	11.74	0.000	Calcium Signaling
TACR2	Calcium Signaling	14.21	0.000	Calcium Signaling
PPIF	Calcium Signaling	4.92	0.000	Calcium Signaling
SPHK2	Calcium Signaling/Food intake	3.18	0.025	Calcium Signaling
GRIN1	Calcium Signaling/Food intake	3.60	0.016	Calcium Signaling
FOXA2	Food intake	19.37	0.000	NA
SDC1	Food intake	10.74	0.000	NA
AGT	Leptin Response/Food Intake/FOXO1	4.23	0.029	NA
IRX5	Leptin response/Food intake/Neurogenesis	3.83	0.006	NA
GRB2	Leptin Response	3.36	0.003	NA
FOXO4	Leptin Response	5.99	0.000	NA
GPR3	Thermogenesis/Obesity	3.77	0.002	NA
KCNJ10	Metabolic homeostasis/Tanycyte formation	4.87	0.002	NA
ELOVL2	Lipid Metabolism/Elongation of VLFA	6.29	0.001	NA
MRPS6	Mitochondria protein synthesis/Parkinson's	4.00	0.000	NA
ATG101	Autophagy	3.95	0.000	Autophagy
ATG4D	Autophagy	3.84	0.000	Autophagy
MLST8	Autophagy/mTor pathway	8.03	0.000	Autophagy
LAMTOR3	Modulates mTor pathway	5.37	0.000	NA
SIRT7	Neuroprotective during neurogenesis	4.64	0.004	NA
HTRA2	Aging/Cell and organ size/Neuroprotection	3.86	0.000	NA
CCDC22	NF-KB Regulation/Ritscher-Schinzel	6.95	0.000	NA
RGS14	Suppressed synaptic plasticity (LTP)	6.21	0.002	NA
DLGAP3	OCD	5.72	0.001	NA
EMC8	Protein homeostasis of GABAergic neurons/ER membrane complex	3.42	0.021	NA
RGS4	GABAergic/Photoperiod/Environmental processing	6.81	0.002	NA
RBFOX3	Promotes sleep/Associated with epilepsy	7.10	0.000	NA
B4GALNT1	Ganglioside synthesis/Promotes BACE1	4.30	0.008	NA

Table 1

Significant shrew-specific upregulation of genes associated with calcium signaling pathways, blood brain barrier plasticity, food intake and leptin response, and other related functions.

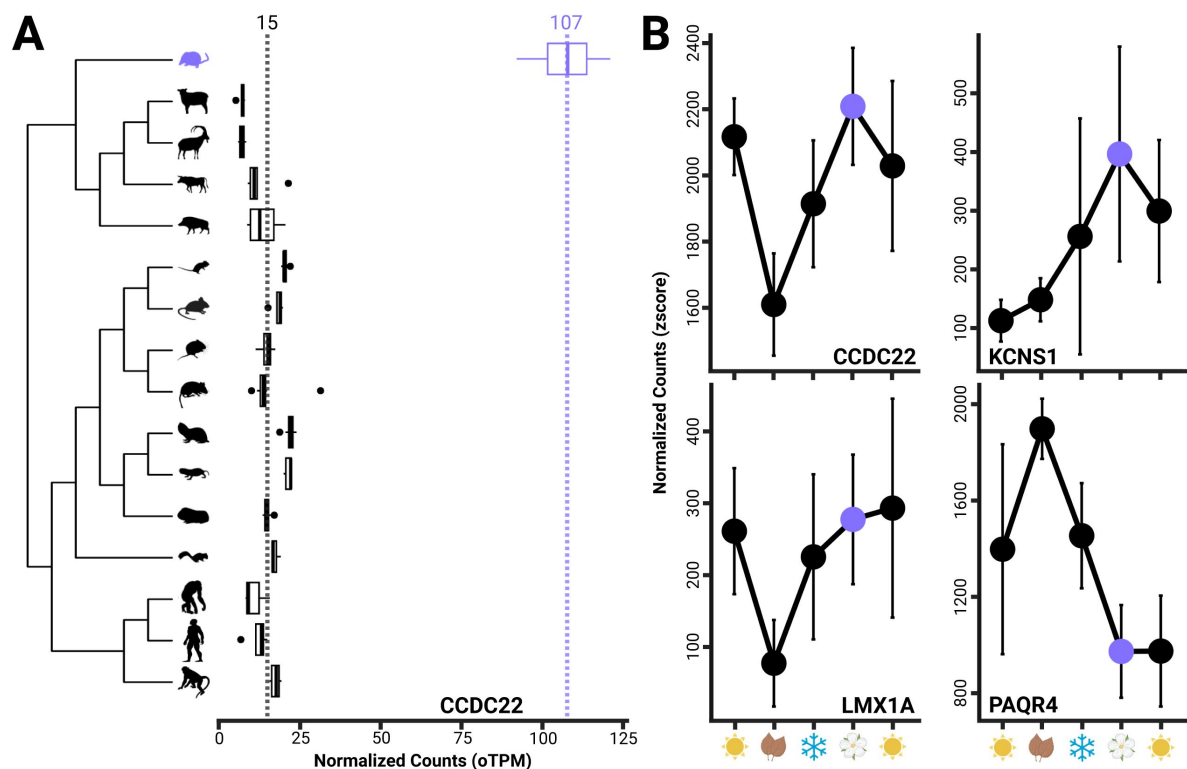


Figure 4

Boxplots of *CCDC22* showing both evolutionary upregulation (**A**) in the shrew and differential expression between spring and autumn individuals, as found in three other genes (*KCNS1*, *LMX1A*, *PAQR4*) (**B**), which have been implicated in the development and progression of human neurological disorders.

analyses identified a suite of genes related to energy homeostasis and environmental sensing that were either seasonally plastic or upregulated in the shrew compared to other mammals, reinforcing the profound role of metabolism in the evolution and regulation of Dehnel's phenomenon. We also found seasonal changes in genes associated with cancer pathways and apoptosis, indicating that the regulation of cell proliferation and death is critical during brain size change. Notably, some of these results resemble expression changes found in human neurological and metabolic disease, which may prove important for future research on therapeutic treatments.

Seasonal plasticity associated with maintaining metabolic homeostasis

Dehnel's phenomenon has been proposed to permit winter activity by reducing shrew energy requirements by limiting energy devoted to the maintenance of larger tissues (4 [↗](#), 7 [↗](#), 20 [↗](#)). This suggests that the maintenance of homeostasis, especially processes related to energy balance, is critical during Dehnel's phenomenon. Seasonal hypothalamus expression data corroborate this hypothesis, as we found three pathways enriched with seasonally varying genes in the hypothalamus with metabolic associations: relaxin signaling, phosphoinositide-3 kinase (PI3K) signaling, and neuron ligand receptor interactions. Relaxin, a neuropeptide, plays a pivotal role in many physiological functions in model organisms, including food and water intake (47 [↗](#)–50 [↗](#)); food restriction upregulates relaxin-3 (*RLN3*) *in vivo* (51 [↗](#)), while administering RLN3 protein into the rat hypothalamus vastly increases food intake (52 [↗](#)–55 [↗](#)). Relaxin has also been found to activate the PI3K pathway (56 [↗](#), 57 [↗](#)), which mediates feeding behavior and glucose homeostasis (58 [↗](#)–62 [↗](#)), as well as angiogenesis via vascular endothelial growth factor (VEGF) production (57 [↗](#), 63 [↗](#), 64 [↗](#)). PI3Ks activity influences feeding behavior and weight gain in response to key metabolic hormones, including leptin and insulin (60 [↗](#)–62 [↗](#), 65 [↗](#)). Lastly, the neuroactive ligand-receptor interaction pathway, a subclass of Kyoto Encyclopedia of Genes and Genomes (KEGG) Environmental Information Processing pathway, is associated with responses in model organisms to shifts in diet (66 [↗](#)), stress (67 [↗](#)), and temperature (68 [↗](#), 69 [↗](#)), all of which the shrew experiences during seasonal changes. Although exploration of gene function in *S. araneus* has only begun, seasonal expression in the hypothalamus highlights that while remaining active in winter with high metabolic rates (20 [↗](#)), shrews actively regulate feeding behavior, food intake, and energy homeostasis.

Putative hypothalamus adaptations may enhance regulatory responses

Sorex araneus have disproportionately high metabolic rates (20 [↗](#)), which they must maintain in winter. While seasonal transcriptomics indicate active regulation of processes related to energy homeostasis, we hypothesized that natural selection may select for effective strategies in comparison to other mammals. To test this, we used comparative analyses which indicated shrews may have evolved adaptive environmental sensing. We found upregulation of several proangiogenic genes associated with the formation and function of the blood brain barrier, including *VEGFA* and *WNT7A*. In normal neural development, neural progenitor cells and surrounding astro- and pericytes express these two genes defining an expression gradient that guides blood vessels to form the blood brain barrier (70 [↗](#)–72 [↗](#)).

While this suggests *VEGFA* and *WNT7A* upregulation in the shrew may increase vascularization of the brain, continued expression of *VEGFA* beyond development increases BBB permeability, leading to a breakdown of its protective effects (73 [↗](#), 74 [↗](#)). BBB permeability is a common symptom of many disease-like states including neuroinflammation (75 [↗](#)) and neurodegeneration (75 [↗](#)–77 [↗](#)). The hypothalamus BBB consists of tanycytes and highly fenestrated capillaries with less tight junctions (78 [↗](#)), replacing the less permeable BBB of other brain regions with a specialized barrier that improves nutrient and energy sensing through dynamic passage of molecules from peripheral circulation (79 [↗](#)–81 [↗](#)). Experiments on mice indicate reduced blood

glucose levels from fasting promote capillary fenestration in the hypothalamic BBB via upregulation of *VEGFA*, eliciting feeding responses through increased hypothalamic exposure to glucose levels (79 [↗](#)). In shrews, upregulation of *VEGFA* and *WNT7A* may constitutively increase capillary fenestration, a potential adaptation of the BBB to improve hypothalamus metabolic sensing and signaling in response to high energy demands. Notably, adaptive expression of *VEGFA* has also been identified in the seasonal vascular plasticity of other mammal species, including reproductive sheep (82 [↗](#), 83 [↗](#)), bison (84 [↗](#)), and squirrels (85 [↗](#)), thus shrew upregulation of *VEGFA* to meet unusually high metabolic demands may parallel mechanisms used seasonally in reproduction.

We also found evidence for adaptive expression relating to maintenance of energy balance, specifically in feeding behavior and leptin responses. Key genes associated with the hypothalamic leptin response, such as *NPY*, *AgRP*, or *POMC*, were not universally orthologous and so could not be analyzed, but there was evolutionary upregulation of many genes involved in downstream leptin responses (*FOXO4*, *GRB2*, *IRX5*, *AGT*) (Figure 3 [↗](#), Table 1 [↗](#)). Unlike their upstream effectors, these genes may have subtle effects on continuous feeding behavior. For example, in the hypothalamus *FOXO1* negatively regulates *POMC* and promotes *AgRP* expression (86 [↗](#)), which can contribute to leptin resistance resulting in a long term appetite suppression (87 [↗](#)–89 [↗](#)). However, much less is known about the hypothalamic functions of the *FOXO1* paralog, *FOXO4*. In mouse adipocytes, leptin-dependent expression of *FOXO4* can rapidly clear blood glucose levels (90 [↗](#)), which can lead to energy storage and decreased satiety. We hypothesize that, in shrews, upregulation of *FOXO4* could either increase blood glucose clearance or leptin resistance, reducing long term satiation and promoting continuous feeding. Functionally, evolutionary upregulation of these genes in the shrew hypothalamus may result in evolved leptin insensitivity, instead of the development of leptin resistance. Reduced leptin sensitivity in shrews could reduce overall satiation, promoting foraging even when the short-term energy budget is balanced thereby improving anticipatory storage before winter.

Lastly, calcium signaling is a ubiquitous form of neural communication (91 [↗](#), 92 [↗](#)) that occurs through the regulation of intra and extracellular calcium concentrations, transmitting signals that can release neurotransmitters (93 [↗](#), 94 [↗](#)) and promote gene expression (95 [↗](#)–98 [↗](#)). Thus, it has been implicated in numerous functions including dendrite growth (99 [↗](#), 100 [↗](#)), synaptogenesis (101 [↗](#)–103 [↗](#)), and initiation and maintenance of long-term potentiation (104 [↗](#), 105 [↗](#)). In shrews, adaptive calcium signaling through the upregulation of several calcium-responsive genes (*VEGFA*, *CASQ1*, *HRH2*, *TACR2*, *PPIF*, *SPHK2*, *GRIN1*) may improve environmental information relay in the hypothalamus, both internally and throughout the central nervous system.

Evolution of brain size plasticity: roles of cellular proliferation, death, and links to neurological disease

Progressive cell loss in neurons is a hallmark of neurodegeneration (106 [↗](#)), but despite seasonal decreases in brain mass and volume, shrews show no reduction in cell number (107 [↗](#)). Uncorrelated changes in size and neuron count in shrews might suggest they evolved neuroprotective mechanisms to reduce cell loss associated with decreased brain size. We found an enrichment of genes upregulated in autumn shrews involved in cancer pathways and apoptosis regulation. The processes underlying cancer and neurodegeneration oppose one another, with increased resistance to cell death driving cancer and premature cell death driving neurodegeneration. Shrews appear to traverse this tightrope by regulating these pathways to avoid both the cell loss typical of neurodegeneration and unchecked proliferation of cancer. This may represent a unique neuroprotective mechanism.

BCL2L1-mediated apoptosis seems to play a critical role in helping shrews maintain this balance during seasonal size change. As a known anti-apoptotic gene, *BCL2L1* promotes cell survival by reducing apoptosis through an NFκB-dependent pathway, as demonstrated in rodents (108–110). The BCL2 gene family is also recognized for its contrasting roles in disease, with upregulation in central nervous system disorders and downregulation in cancers. We found an enrichment of the apoptosis pathway with upregulated genes in autumn, including *BCL2L1*, which may explain why shrews do not exhibit the neuronal loss typically associated with human neurodegeneration. This suggests that the mechanism parallels aspects of neurodegenerative disease without the associated pathology, which we propose could be another neuroprotective mechanism to reduce cell death during brain shrinkage in shrews.

The regulation of apoptosis may represent a convergently evolved strategy across species managing environmental and metabolic stress. For example, natural experiments in hibernators found seasonal reduction of apoptosis, paired with upregulation of *BCL2L1* (111, 112), including in the brain (113). Specific to Dehnel's phenomenon, *BCL2L1* upregulation in autumn indicates tight regulation of cell survival during brain size changes and periods of metabolic and environmental stress: a balance between brain shrinkage and maintenance of cell numbers. Evolutionary change in shrew gene expression likely benefits this cellular regulation. We found an evolutionary upregulation of genes associated with autophagy, which is the degradation of cellular parts to provide energy and materials to maintain cellular homeostasis under stressful conditions, in the shrew hypothalamus. This included *MLST8*, an integral component of the mTOR protein complex 1 (mTORC1) that regulates sugar and lipid metabolism (114), but also has recently been found to modulate autophagy in response to nutrient deprivation (115). Two evolutionarily upregulated genes *PPIF* and *SPHK2*, can also promote apoptosis associated with mitochondrial calcium influx. *PPIF* is an important component of the mitochondrial permeability transition, which responds to cellular calcium overload and oxidative stress by promoting cell death (116–119). Similar to *PPIF*, overexpression of *SPHK2* in stressed cells induces cell cycle arrest and promotes apoptosis through a series of signaling cascades that increase cytosolic calcium and promote the release of pro-apoptotic cytochrome C from the mitochondria (120, 121). While both *SPHK2* and *PPIF* are overexpressed and likely contribute to neuronal death in both human patients with Huntington's disease or Alzheimer's disease and rodent models of these diseases, knockout or inhibition of these genes protects against neurodegeneration (122–125). In the shrew hypothalamus, canalized upregulation of these genes likely influences cell survival and death, but precisely how requires further investigation.

Lastly, *CCDC22*, *KCNS1*, *LMX1A*, and *PAQR4*, are both seasonally plastic and evolutionarily upregulated in the shrew hypothalamus (Figure 4). While the number of overlapping genes is not significantly greater than expected by chance, the combination of seasonal and evolutionary upregulation suggests these genes may play integral roles in Dehnel's phenomenon. Notably, some of these genes are directly related to the development and progression of neurological disorders in humans. For example, *CCDC22* regulation in shrews may provide protection against harm incurred during seasonal size changes. The coiled-coil domain containing 22 protein (*CCDC22*) plays a key role in endosomal recycling of proteins and is a novel candidate gene for several neurological disorders (126–129). *CCDC22* is a subunit of the CCC (*CCDC22*, *CCDC93*, *COMMD*) complex that aids in trafficking and recycling of endosomal membrane proteins (43, 130, 131). Improper recycling and resulting protein aggregation readily occurs in neurons, as they do not proliferate (132, 133), and has also been implicated in neurodegenerative diseases, including cytotoxicity in amyotrophic lateral sclerosis (134), amyloid-β aggregation in Alzheimer's disease (135–137), and *MUL1* degradation in Parkinson's disease (138). Missense mutations and subsequent downregulation of *CCDC22*, are potential mechanisms underlying X-linked intellectual disability (126, 127) and Ritscher-Schinzel syndrome (127–129) in humans. Both disorders are characterized by macrocephaly, malformations of the brain, craniofacial abnormalities, and intellectual disabilities. In pubescent spring shrews, we found *CCDC22* has a ~7-fold upregulation compared to other mammals and a significant upregulation

during spring compared to autumn shrinkage. Neuroprotective functions may be especially important in spring, as brain mass returns increasing the need for endosomal recycling of proteins and may explain seasonal plasticity in *CCDC22* expression in tandem with adaptive upregulation.

Limitations

Despite the proposed adaptive roles of the many genes discussed in our study, several limitations must be acknowledged. During shrew seasonal size plasticity, shrew environments may differ from those of the other species included in this experiment. These environmental differences could influence gene expression, leading to over- or underestimation of selection in the shrew lineage. Although we controlled for several sources of variation by using only pubescent and wild or non-domesticated individuals when feasible, other sources of variation remain unaccounted for because of practical constraints. The functions we propose for many genes remain speculative and require further validation. For example, we tested the hypothesis that autumn upregulation of *BCL2L1* reduces apoptosis by overexpressing this gene in a related mammalian neural cell line. However, this experiment did not demonstrate a rescue effect against heat-induced apoptosis, contrary to our expectations. While this unexpected result could stem from several causes, such as the anti-apoptotic function of *BCL2L1* being specific to the unique neural environment of shrews or arise through species-specific gene interactions, it may also reflect limitations in our study, highlighting the need for further investigation to validate and refine our conclusions.

Conclusion

We discovered evidence for adaptive evolution of gene expression involved in the regulation of metabolism, cell survival, and size plasticity associated with Dehnel's phenomenon. Not only does the hypothalamus shrink to reduce energetic load during winter, but it must also control metabolic homeostasis during these changes. Our results reveal both conserved and novel gene expression mechanisms that likely underlie the central functions of the hypothalamus during seasonal size plasticity. While previous work had hinted at the metabolic nature of Dehnel's phenomenon, by pairing seasonal differential expression with comparisons to other species, we identified adaptations likely involved in the evolution of this unique brain size plasticity. Metabolism, body size, and longevity are intrinsically linked life history factors. In the shrew, unusually high metabolic rates force winter activity, leading to intense metabolic sensing and regulation, changes in cell survival pathways, and –through both the evolution of Dehnel's phenomenon and these sensing and regulatory adaptations– shortened lifespans. Together, these gene expression adaptations likely underlie drastic seasonal size change and mitigate its detrimental effects.

Materials and Methods

Shrew Sample Collection

We used tissues from shrews collected for experiments analyzing gene expression and metabolic shifts in brain regions through Dehnel's phenomenon ([12](#)). Briefly, *S. araneus* were collected from a single German population (47.9684N, 8.9761 E) across five different seasons of Dehnel's phenomenon from June 2020-June 2021 ($n=25$, $n_{\text{per_season}} = 4-5$). Shrews were trapped with insulated wooden traps that contained mealworms, which were checked every two hours. This protocol was used to minimize trap-related stress from heat/cold shock or lack of food and thus reduce stress-related variation on gene expression. Shrews were then euthanized via vascular perfusion of PAXgene Tissue Fixative. Brain regions were dissected into individual regions in cold fixative and then incubated in PAXgene Tissue Stabilizer for 2-24 hours before long-term storage. Samples were preserved in stabilizer at -180°C in liquid nitrogen until RNA extraction.

RNA Extraction, Library Preparation, and Sequencing

We extracted the *S. araneus* hypothalamus RNA using a modified Qiagen Micro RNAeasy protocol used for small mammalian sensory tissues (139 [↗](#)). These extractions reduce degradation due to heat and improve RIN from standard Qiagen Micro RNAeasy protocols by disrupting tissue on dry ice using glass, instead of plastic, mortar and pestles. RNA was sent to Azenta Life Sciences for quality control (nanodrop and RNA ScreenTape), library preparation, and sequencing. Hypothalamus libraries were prepared with poly-A selection and sequenced for approximately 15–25 million reads per sample in 150bp PE reads.

RNA sequences from other species were collected from the NIH National Center for Biotechnology Information's (NCBI) Sequence Read Archive data sets. First, we searched the Sequencing Read Archives (SRA) for the keyword hypothalamus, filtered by both RNA and mammal species. As models described below rely on using the variance both between and within species, species required three or more biological samples to be included. This removed any species that had only one hypothalamus sample sequenced. Species were excluded from the study if a genome assembly was not readily available (*Phodopus sungorus*), as this information was needed for an unbiased alignment of reads. Our final criterion was to only include species for which post-pubescent individuals were available to reduce noise from age effects and the onset of puberty. If the onset of puberty was not specifically stated for the dataset, AnAge (140 [↗](#)) was used to determine if samples were pubescent. *Oryctolagus cuniculus* and *Pan paniscus* were prepubescent and thus removed. For the remaining species, if multiple hypothalamus RNA-seq data sets were available, we used less stringent rules to determine which data to select. First, to reduce domestication or captivity effects, we chose wild or non-domesticated individuals over laboratory-raised animals. Second, to reduce the variance from different extraction protocols and sequencing methods, we used samples from the same experiments or larger data sets, available for multiple species or brain regions. Using samples from datasets with multiple regions or species also suggests intimate knowledge of neural anatomy, increasing confidence in dissections. Third, we used a maximum of eight samples per species and attempted to maintain a 1:1 sex ratio when possible. We did not filter by sequencing method (e.g., used 50bp SE as well as 150 PE reads) or sequencing depth, as filtering poor reads and normalizing for library depth and content should account for both these factors.

RNA Quantification, Normalization, and Orthology

Adapter sequences were trimmed from the raw reads using the default parameters of fastp (141 [↗](#)), which is able to autodetect adapters regardless of different library preparations across species. This program also corrected and pruned low quality bases and reads using a sliding window approach. Processed reads were aligned and quantified with Kallisto 0.46.2 (142 [↗](#)), which probabilistically estimates gene counts through pseudo-alignment to each species-specific genome assembly. Samples were removed from the data set if mapping rate was below 30%, indicating either poor sequencing or low assembly quality. Additionally, novel shrew hypothalamus sequences were compared against previously published shrew region data to verify tissue type. Gene counts for all species were then normalized for total library size into Transcripts Per Million (TPM). Orthologous genes between species were inferred with OrthoFinder (143 [↗](#)) using default parameters, retaining only single-copy orthologs identified across all species ($n=6,496$ single-copy orthologs). Visualizations of the frequency distribution of the orthologous (oTPMs; Supplemental Figure 1) were used to identify outlier species. For example, *Papio anubis* was removed from this experiment because of its extremely low mapping percentage (mean 16.75%; range 9.5%–23.7%), which had a large effect on the oTPM distribution.

Branch-shift Changes in Expression

We modeled gene expression as a trait using phylogenetic comparative methods, as these account for evolutionary divergence among species (144 [↗](#)). We ran Expression Variance and Evolution models (EVE) to test for significant changes of expression in the shrew lineage, using post pubescent spring individuals (34 [↗](#)). EVE models both parameterize and estimate the ratio (β) of population (within species) to evolutionary (between species) variance, such that high β ratios indicate expression plasticity, and low β ratios indicate differential expression between species, necessary for inferring selection. We ran two EVE models to test for divergence in expression in the shrew lineage. First, we tested for branch-specific expression level shifts by contrasting the likelihood of two Ornstein-Uhlenbeck models for the data, one with a single expression optimum (null; stabilizing selection), and another with a second optimum on the shrew branch (selection). A likelihood ratio test between the null and selective hypotheses for each gene (X^2) was conducted using these likelihoods. A Bonferroni correction was used to account for multiple hypothesis tests to identify candidate genes under selection ($p_{adj} < .05$). For the second model, we ran a dropout test to further validate the specificity of the expression change in the shrew lineage. After removing the shrew expression data, we identified genes with high expression divergence across the phylogeny (significantly low β ratios), which would indicate relaxed or diversifying selection. This gene list was compared to that from first model to prune genes not specifically associated with divergence in the shrew. Both models described above ran on the Bayesian molecular-clock mammalian phylogeny of (145 [↗](#)) pruned to match our species sample. Finally, we used the DAVID functional annotation tool (146 [↗](#)) to determine enriched KEGG Gene Ontology (GO) pathways from the candidate gene set.

Temporal Clustering

To analyze the temporal variation of shrew expression in the hypothalamus, we temporally clustered our data using the package TCseq (147 [↗](#)). We began by further normalizing our data across Dehnel's phenomenon seasons using the median of ratios from DESeq2 (148 [↗](#)), which normalized for library size and content. Genes with consistent expression across time with little variation between seasons would not be associated with observed phenotypic changes. These genes were filtered from this data set by only selecting genes with an absolute fold change of 0.5 between any two seasons. We also removed genes with low expression that would appear to have high fold changes despite lacking enough transcript expression to influence phenotype, retaining only reads with two samples > 10 normalized reads. After filtering, counts for each season were converted to mean z-scores and then clustered into groups of similar gene expression profiles using fuzzy cmeans clustering. The number of resulting gene clusters was calculated a priori, by bootstrapping ($n=20$) a gap statistic that minimizes within-cluster distances. Genes with highest membership in clusters associated with variation across all seasons (seasonal effect; Clusters 1, 4, 5, 6, 7, 9, 10), as compared to those that just show extensive differences (>1.25 standard deviations) between the summer seasons (juvenile and adult) without a second local maximum ($\text{abs}(\text{autumn-winter}) < 0.5$ and $\text{abs}(\text{winter-summer adult}) < 0.5$) (developmental effect; Clusters 2, 3, 8, 11, 12), were analyzed for KEGG GO enrichment using DAVID functional annotation (146 [↗](#)). Additionally, each cluster was analyzed individually and included in the Supplementary Material.

Differential Gene Expression

We tested for differential gene expression between autumn and spring individuals, as these seasons differ in hypothalamic size phenotype (shrinkage vs. regrowth) (38 [↗](#)), and were previously identified divergence in liver expression related to metabolism (12 [↗](#)). To test for significant differential expression, we fit a negative binomial generalized linear model to the normalized (median of ratios) gene counts using DESeq2 (148 [↗](#)). We then tested for significant differences in gene expression between autumn and spring using a Wald test, followed by multiple testing correction on resulting p-values with the Benjamini and Hochberg procedure (149 [↗](#)). Sex

was used as an additional covariate (~sex + condition), but our design is imbalanced as not every season samples both sexes. Significantly differentially expressed genes were binned by fold change (>1.58 log-fold change), to quantify differentially expressed genes of high effect. Significant differentially expressed genes were also used to identify KEGG GO enrichment ([146](#)), and compared against the candidate list of genes with branch shift changes in expression to identify genes both plastic across seasons and consistent with expression adaptation in the shrew. To test if observed overlap was greater than expected by chance, we performed a Fisher's exact test.

Cell culture

Mustelo putoris furo (MPF) brain cells (ATCC, CRL-1656) were cultured in 10mL of Basal Medium Eagle (BME) supplemented with 14% sheep serum. Cells were cultured in 60.8-cm² treated tissue culture dishes at 37°C in 5% CO₂ atmosphere and 95% humidity. With induced cell death, cells were seeded into treated flat bottom six-well plates with 2mL/well of complete culture media.

Cell death induction

Three experiments were conducted to induce cell death in MPF cells (cold induced, peroxide induced, and heat induced), with only heat causing significant decrease in cell viability. *Cold induced*: Two 60.8-cm² tissue culture dishes of MPF cells were seeded into two six-well plates for 24 hours before reaching 80% confluency. In one experiment, "short cold" apoptosis was performed by immersing the six-well plate into a 0°C ice-water bath for 2 hours, followed by 3 hours of rewarming to 37°C in an incubator. Cells were then treated with crystal violet staining. In a similar experiment, "long cold" apoptosis was performed by immersing the six-well plate into a 0°C water bath for 4 hours, followed by 24 hours of rewarming to 37°C before crystal violet staining. *Heat-induced cell death*: Two 60.8-cm² tissue culture dishes of MPF cells were seeded into two six-well plates for 24 hours before reaching 80% confluency. Heat cell death was attempted by immersing the six-well plate into a 45°C water bath for 2 hours, followed by re-cooling to 37°C prior to crystal violet staining.

In vitro transcription and transfection of cells

The AmpliScribe T7 High Yield Transcription kit (Lucigen) was used for RNA in vitro transcription with a 2-hour incubation at 37°C using gblock sequences available in supplemental. BCL2L1 sequences were designed with a T7 promoter inserted at the beginning to facilitate transcription (Integrated DNA Technologies). We selected *BCL2L1* because it is the gene furthest downstream in the apoptotic pathway, thus making it the most directly involved gene in programmed cell death, whereas upstream genes could influence additional genes or alternative processes. RNA transcripts were purified using the Qiagen miRNA kit and concentration determined using a Nanodrop2000 (Thermo Fisher Scientific). Two 60.8-cm² tissue culture dishes of MPF cells were seeded into two six-well plates for 24 hours before reaching 80% confluency. 300ng of *BCL2L1* RNA or a scrambled form of *BCL2L1* RNA were transfected with Lipofectamine 3000 into each well for 24 hours according to manufacturer's instructions (Thermo Fisher Scientific). In short, two tubes were used to make a master mix for each RNA sample. Tube 1 contained 125μL/well of Opti-mem (Thermo Fisher Scientific) with 7.5 μL/well of Lipo3000. Tube 2 contained 125μL/well of Opti-mem, 5μL/well of P3000 and 300ng/well of RNA. Tube 2 was added to tube 1 and incubated 15 minutes at room temperature. The solution was then gently mixed and dispersed evenly to cells.

Crystal violet cell viability assays

Crystal violet solution was made using 500mg crystal violet powder in 100mL of 50% methanol. Cells were cultured to 80% confluency in two six-well plates. Complete media was aspirated in wells and washed three times in 1mL of 1X DPBS before 500μL of crystal violet solution was added to each well. Plates were wrapped in foil and placed on a shaker for 30 minutes. After time elapsed, crystal violet was removed, and cells were washed with tap water until free color was no longer visible. Plates were left at room temperature for 10-15 minutes until dry. 500μL of 100%

methanol was added to each well and plates were put on a shaker for one hour at room temperature. 100µL of solution was taken from each well in triplicate and added to a 96-well plate. 100µL of 100% methanol was added in triplicate to a row for normalization of background absorbance. Wells were read at 570nm using a spectrophotometer (BioTek). For viability, data are reported as a percentage of control or optical density for absorbance. For statistical analysis, differences were judged to be statistically significant when $p < 0.05$ by a Welch's two sample t-test.

Data Availability

Supplementary tables, data, results, and code deposited and found on Github https://github.com/wrthomas315/Sorex_Hypothalamus_Transcriptomics2. Raw sequencing data located in the NCBI Sequencing Read Archive (BioProject PRJNA941271). Supplemental Figures. S1–S2: DOI: <https://doi.org/10.6084/m9.figshare.26049739.v1>.

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Additional files

Supplemental Data [↗](#)

Supplemental Figure 1 [↗](#)

Supplemental Figure 2 [↗](#)

Supplemental Table 1 [↗](#)

Supplemental Table 2 [↗](#)

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Reviewer #1 (Public review):**Summary:**

In this paper, Thomas et al. set out to study seasonal brain gene expression changes in the Eurasian common shrew. This mammalian species is unusual in that it does not hibernate or migrate but instead stays active all winter while shrinking and then regrowing its brain and other organs. The authors previously examined gene expression changes in two brain regions and the liver. Here, they added data from the hypothalamus, a brain region involved in the regulation of metabolism and homeostasis. The specific goals were to identify genes and gene groups that change expression with the seasons and to identify genes with unusual expression compared to other mammalian species. The reason for this second goal is that genes that change with the season could be due to plastic gene regulation, where the organism simply reacts to environmental change using processes available to all mammals. Such changes are not necessarily indicative of adaptation in the shrew. However, if the same genes are also expression outliers compared to other species that do not show this overwintering strategy, it is more likely that they reflect adaptive changes that contribute to the shrew's unique traits.

The authors succeeded in implementing their experimental design and identified significant genes in each of their specific goals. There was an overlap between these gene lists. The authors provide extensive discussion of the genes they found.

The scope of this paper is quite narrow, as it adds gene expression data for only one additional tissue compared to the authors' previous work in a 2023 preprint. The two papers even use the same animals, which had been collected for that earlier work. As a consequence, the current paper is limited in the results it can present. This is somewhat compensated by an expansive interpretation of the results in the discussion section, but I felt that much of this was too speculative. More importantly, there are several limitations to the design, making it hard to draw stronger conclusions from the data. The main contribution of this work lies in the generated data and the formulation of hypotheses to be tested by future work.

Strengths:

The unique biological model system under study is fascinating. The data were collected in a technically sound manner, and the analyses were done well. The paper is overall very clear, well-written, and easy to follow. It does a thorough job of exploring patterns and enrichments in the various gene sets that are identified.

I specifically applaud the authors for doing a functional follow-up experiment on one of the differentially expressed genes (BCL2L1), even if the results did not support the hypothesis. It is important to report experiments like this and it is terrific to see it done here.

Comments on revised version:

This updated version of the paper is improved compared to its initial version. As such, the strengths remain the same as before, with a fascinating model system and an interesting research question. The earlier weaknesses related to overinterpretation of the data have been largely fixed by shortening the paper and adding appropriate caveats throughout. The paper now also includes a significance test for its overlap between gene lists. While this turned out to be negative (i.e., there is not more overlap between lists than expected by chance), reporting this result transparently has strengthened the paper.

<https://doi.org/10.7554/eLife.100788.2.sa3>

Reviewer #2 (Public review):

Summary:

Shrews go through winter by shrinking their brain and most organs, then regrow them in the spring. The gene expression changes underlying this unusual brain size plasticity were unknown. Here, the authors looked for potential adaptations underlying this trait by looking at differential expression in the hypothalamus. They found enrichments for DE in genes related to the blood brain barrier and calcium signaling, as well as used comparative data to look at gene expression differences that are unique in shrews. This study leverages a fascinating organismal trait to understand plasticity and what might be driving it at the level of gene expression. This manuscript also lays the groundwork for further developing this interesting system.

Strengths:

One strength is that the authors used OU models to look for adaptation in gene expression. The authors also added cell culture work to bolster their findings.

Comments on revised version:

I think that the authors have made a strong revision. No other comments.

<https://doi.org/10.7554/eLife.100788.2.sa2>

Reviewer #3 (Public review):

Summary:

In their study, the authors combine seasonal and comparative transcriptomics to identify candidate genes with plastic, canalized, or lineage-specific (i.e., divergent) expression patterns associated with an unusual overwintering phenomenon (Dehnel's phenomenon - seasonal size plasticity) in the Eurasian shrew. Their focus is on the shrinkage and regrowth of the hypothalamus, a brain region that undergoes significant seasonal size changes in shrews and plays a key role in regulating metabolic homeostasis. Through comparative transcriptomic analysis, they identify genes showing derived (lineage-specific), plastic (seasonally regulated), and canalized (both lineage-specific and plastic) expression patterns. The authors hypothesize that genes involved in pathways such as the blood-brain barrier, metabolic state sensing, and ion-dependent signaling will be enriched among those with notable transcriptomic patterns. They complement their transcriptomic findings with a cell culture-based functional assessment of a candidate gene believed to reduce apoptosis.

Strengths:

The study's rationale and its integration of seasonal and comparative transcriptomics are well-articulated and represent an advancement in the field. The transcriptome, known for its dynamic and plastic nature, is also influenced by evolutionary history. The authors effectively demonstrate how multiple signals-evolutionary, constitutive, and plastic-can be extracted, quantified, and interpreted. The chosen phenotype and study system are particularly compelling, as it not only exemplifies an extreme case of Dehnel's phenotype, but the metabolic requirements of the shrew suggest that genes regulating metabolic homeostasis are under strong selection.

Weaknesses:

The results of the expression patterns are quite compelling and a number of interesting downstream hypotheses are outlined; however, the interpretation of the role of each gene and pathway identified is speculative which dampens the overall impact of the work. That said, I commend the authors on functionally testing one of the differentially expressed genes. I also commend the inclusion of that negative result.

<https://doi.org/10.7554/eLife.100788.2.sa1>

Author response:

The following is the authors' response to the original reviews

eLife Assessment

This study presents valuable findings related to seasonal brain size plasticity in the Eurasian common shrew (Sorex araneus), which is an excellent model system for these studies. The evidence supporting the authors' claims is convincing. However, the authors should be careful when applying the term adaptive to the gene expression changes they observe; it would be challenging to demonstrate the differential fitness effects of these gene expression changes. The work will be of interest to biologists working on neuroscience, plasticity, and evolution.

We appreciate the reviewers' suggestions and comments. For the phylogenetic ANOVA we used (EVE), which tests for a separate RNA expression optimum specific to the shrew lineage consistent with expectations for adaptive evolution of gene expression. But, as you noted, while this analysis highlights many candidate genes evolving in a manner consistent with positive selection, further functional validation is required to confirm if and how these genes contribute to Dehnel's phenomenon. In the discussion, we now emphasize that inferred adaptive expression of these genes is putative and outline that future studies are needed to test the function of proposed adaptations. For example, cell line validations of BCL2L1 on apoptosis is a case study that tests the function of a putatively adaptive change in gene expression, and it illuminates this limitation. We also have refined our discussion to focus more on pathway-level analyses rather than on individual genes, and have addressed other issues presented, including clarity of methods and using sex as a covariate in our analyses.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this paper, Thomas et al. set out to study seasonal brain gene expression changes in the Eurasian common shrew. This mammalian species is unusual in that it does not hibernate or migrate but instead stays active all winter while shrinking and then regrowing its brain and other organs. The authors previously examined gene expression changes in two brain regions and the liver. Here, they added data from the hypothalamus, a brain region involved in the regulation of metabolism and homeostasis. The specific goals were to identify genes and gene groups that change expression with the seasons and to identify genes with unusual expression compared to other mammalian species. The reason for this second goal is that genes that change with the season could be due to plastic gene regulation, where the organism simply reacts to environmental change using processes available to all mammals. Such changes are not necessarily indicative of adaptation in the shrew. However, if the same genes are also expression outliers compared to other species that do not show this overwintering

strategy, it is more likely that they reflect adaptive changes that contribute to the shrew's unique traits.

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Thank you for your interest in our manuscript and for your insights. We addressed your comments below: we now highlight the limitations of our study design in the discussion and emphasize that, while a second optimum of gene expression in shrews is consistent with adaptive evolution, we recognize that not all sources of variation in gene expression can be fully accounted for. We highlight the putative nature of these results in our revisions, especially in our new limitations section (lines 541-555).

Strengths:

The unique biological model system under study is fascinating. The data were collected in a technically sound manner, and the analyses were done well. The paper is overall very clear, well-written, and easy to follow. It does a thorough job of exploring patterns and enrichments in the various gene sets that are identified.

I specifically applaud the authors for doing a functional follow-up experiment on one of the differentially expressed genes (BCL2L1), even if the results did not support the hypothesis. It is important to report experiments like this and it is terrific to see it done here.

We are glad to hear that you found our manuscript fascinating and clearly written. While we hoped to see an effect of BCL2L1 on apoptosis as proposed, we agree that reporting null results is valuable when validating evolutionary inferences.

Weaknesses:

While the paper successfully identifies differentially expressed seasonal genes, the real question is (as explained by the authors) whether these are evolved adaptations in the shrews or whether they reflect plastic changes that also exist in other species. This question was the motivation for the inter-species analyses in the paper, but in my view, these cannot rigorously address this question. Presumably, the data from the other species were not collected in comparable environments as those experienced by the shrews studied here. Instead, they likely (it is not specified, and might not be knowable for the public data) reflect baseline gene expression. To see why this is problematic, consider this analogy: if we were to compare gene expression in the immune system of an individual undergoing an acute infection to other, uninfected individuals, we would see many, strong expression differences. However, it would not be appropriate to claim that the infected individual has unique features - the relevant physiological changes are simply not triggered in the other individuals. The same applies here: it is hard to draw conclusions from seasonal expression data in the shrews to non-seasonal data in the

other species, as shrew outlier genes might still reflect physiological changes that weren't active in the other species.

There is no solution for this design flaw given the public data available to the authors except for creating matched data in the other species, which is of course not feasible. The authors should acknowledge and discuss this shortcoming in the paper.

Thank you for taking the time to provide such insightful feedback. As you noted, while shrews experience seasonal size changes, their environments may differ from the other species used in this experiment, leading to increased or decreased expression of certain genes and reducing our ability accurately detect selection across the phylogeny. Although we sought to control for as many sources of variation as possible, such as using only post-pubescent, wild, or non-domesticated individuals when feasible, we recognize that not all sources of variation can be fully accounted for within a practical experiment. We agree that these sources of variation can introduce both false positives and negatives into our results, and we have now highlighted this limitation within our discussion (lines 538-552).

Related to the point above: in the section "Evolutionary Divergence in Expression" it is not clear which of the shrew samples were used. Was it all of them, or only those from winter, fall, etc? One might expect different results depending on this. E.g., there could be fewer genes with inferred adaptive change when using only summer samples. The authors should specify which samples were included in these analyses, and, if all samples were used, conduct a robustness analysis to see which of their detected genes survive the exclusion of certain time points.

Thank you for this attention to detail. We used spring adults for this analysis. This decision was made as only used post pubescent individuals for all species in the analysis, and this was the only season where adult shrews were going through Dehnel's phenomenon. We have now clarified this in both the methods and results (line 247 and line 667)

In the same section, were there also genes with lower shrew expression? None are mentioned in the text, so did the authors not test for this direction, or did they test and there were no significant hits?

We did test for decreased shrew expression compared to the rest of the species, but there were no significant genes with significant decreases. We hypothesize that there are two potential reasons for this results; 1) If a gene were to be selected for decreased expression, selection for constitutive expression of the gene across all species may be weak, and thus found in other lineages as well, or 2) decreased or no expression may relax selection on the coding regions, and thus these genes are not pulled out as we identify 1:1 orthologs. This is consistent with results provided from the original methods manuscript. Thank you for pointing out that we did not discuss this information in the text, and we now include it in our results (lines 250-251).

The Discussion is too long and detailed, given that it can ultimately only speculate about what the various expression changes might mean. Many of the specific points made (e.g. about the blood-brain-barrier being more permissive to sensing metabolic state, about cross-organ communication, the paragraphs on single, specific genes) are a stretch based on the available data. Illustrating this point, the one follow-up experiment the authors did (on BCL2L1) did not give the expected result. I really applaud the authors for having done this experiment, which goes beyond typical studies in this space. At the same time, its result highlights the dangers of reading too much into differential expression analyses.

We agree with your point, while our extensive discussion is useful for testing future hypotheses, ultimately some of the discussion may be too speculative for our readers. To amend this, we have reduced some portions of our discussion and focused more on pathways than individual genes, including removing mechanisms related to *HRH2*, *FAM57B*, *GPR3*, and GABAergic neurons. We hope that this highlights to the reader the speculative nature of many of our results.

There is no test of whether the five genes observed in both analyses (seasonal change and inter-species) exceed the number expected by chance. When two gene sets are drawn at random, some overlap is expected randomly. The expected overlap can be computed by repeated draws of pairs of random sets of the same size as seen in real data and by noting the overlap between the random pairs. If this random distribution often includes sets of five genes, this weakens the conclusions that can be drawn from the genes observed in the real data.

Thank you for highlighting this approach, it is greatly needed. After running this test, we found that observed overlapping genes were more than the expected overlap, yet not significant. We now show this in our methods (lines 277-278) and results (lines 719-720).

Reviewer #2 (Public review):

Summary:

Shrews go through winter by shrinking their brain and most organs, then regrow them in the spring. The gene expression changes underlying this unusual brain size plasticity were unknown. Here, the authors looked for potential adaptations underlying this trait by looking at differential expression in the hypothalamus. They found enrichments for DE in genes related to the blood-brain barrier and calcium signaling, as well as used comparative data to look at gene expression differences that are unique in shrews. This study leverages a fascinating organismal trait to understand plasticity and what might be driving it at the level of gene expression. This manuscript also lays the groundwork for further developing this interesting system.

We are glad you found our manuscript interesting and thank and thank you for your feedback. We hope that we have addressed all of your concerns as described below.

Strengths:

One strength is that the authors used OU models to look for adaptation in gene expression. The authors also added cell culture work to bolster their findings.

Weaknesses:

I think that there should be a bit more of an introduction to Dehnel's phenomenon, given how much it is used throughout.

Thank you for this insight. With a lengthy introduction and discussion, we agree that the importance of Dehnel's phenomenon may have been overshadowed. We have shortened both sections and emphasized the background on Dehnel's phenomenon in the first two paragraphs of the introduction, allowing this extraordinary seasonal size plasticity to stand out.

Reviewer #3 (Public review):

Summary:

In their study, the authors combine developmental and comparative transcriptomics to identify candidate genes with plastic, canalized, or lineage-specific (i.e., divergent) expression patterns associated with an unusual overwintering phenomenon (Dehnel's phenomenon - seasonal size plasticity) in the Eurasian shrew. Their focus is on the shrinkage and regrowth of the hypothalamus, a brain region that undergoes significant seasonal size changes in shrews and plays a key role in regulating metabolic homeostasis. Through combined transcriptomic analysis, they identify genes showing derived (lineage-specific), plastic (seasonally regulated), and canalized (both lineage-specific and plastic) expression patterns. The authors hypothesize that genes involved in pathways such as the blood-brain barrier, metabolic state sensing, and ion-dependent signaling will be enriched among those with notable transcriptomic patterns. They complement their transcriptomic findings with a cell culture-based functional assessment of a candidate gene believed to reduce apoptosis.

Strengths:

The study's rationale and its integration of developmental and comparative transcriptomics are well-articulated and represent an advancement in the field. The transcriptome, known for its dynamic and plastic nature, is also influenced by evolutionary history. The authors effectively demonstrate how multiple signals-evolutionary, constitutive, and plastic-can be extracted, quantified, and interpreted. The chosen phenotype and study system are particularly compelling, as it not only exemplifies an extreme case of Dehnel's phenotype, but the metabolic requirements of the shrew suggest that genes regulating metabolic homeostasis are under strong selection.

Weaknesses:

(1) In a number of places (described in detail below), the motivation for the experimental, analytical, or visualization approach is unclear and may obscure or prevent discoveries.

Thank you for finding our research and manuscript compelling, as well as the valuable feedback that will drastically improve our manuscript. We hope that we have alleviated your concerns below by following your instructions below.

(2) Temporal Expression - Figure 1 and Supplemental Figure 2 and associated text:

- It is unclear whether quantitative criteria were used to distinguish "developmental shift" clusters from "season shift" clusters. A visual inspection of Supplemental Figure 2 suggests that some clusters (e.g., clusters 2, 8, and to a lesser extent 12) show seasonal variation, not just developmental differences between stages 1 and 2. While clustering helps to visualize expression patterns, it may not be the most appropriate filter in this case, particularly since all "season shift" clusters are later combined in KEGG pathway and GO analyses (Figure 1B).

- The authors do not indicate whether they perform cluster-specific GO or KEGG pathway enrichment analyses. The current analysis picks up relevant pathways for hypothalamic control of homeostasis, which is a useful validation, but this approach might not fully address the study's key hypotheses.

Thank you for this valuable feedback. We did not want to include clusters we deemed to be related to development, as this should not be attributed to changes associated with Dehnel's phenomenon. We did this through qualitative, visual inspection, which we realize can differ between parties (i.e., clusters 2, 8, and 12 appeared to be seasonal). Qualitatively, we were looking for extreme divergence between Stage 1 and Stage 5 individuals, as expression was

related to season and not development, then the average of these stages within cluster should be relatively similar. We have now quantified this as large differences in z-score ($\text{abs}(\text{summer juvenile}-\text{summer adult}) > 1.25$) without meaningful interseason variations determined by a second local maximum ($\text{abs}(\text{autumn}-\text{winter}) < 0.5$ and $\text{abs}(\text{winter}-\text{summer}) < 0.5$), and added it both our methods (lines 699-702) and results (line 192).

Regarding the combination of clusters for pathway enrichment compared to individual pathways, we agree that combining clusters may be more informative for overall homeostasis, compared to individual clusters which may inform us on processes directly related to Dehnel's phenomenon. Initially, we were tentative to conduct this analysis, as clusters contain small gene sets, reducing the ability to detect pathway enrichments. We have now included this analysis, which is reported in our methods (lines 703-704), results (lines 203-204), and new supplemental table.

(3) Differential expression between shrinkage (stage 2) and regrowth (stage 4) and cell culture targets

- The rationale for selecting BCL2L1 for cell culture experiments should be clarified. While it is part of the apoptosis pathway, several other apoptosis-related genes were identified in the differential gene expression (DGE) analysis, some showing stronger differential expression or shrew-specific branch shifts. Why was BCL2L1 prioritized over these other candidates?

We agree that our rationale for validating BCL2L1 function in neural cell lines was not clearly explained in the manuscript. We selected BCL2L1 because it is the furthest downstream gene in the apoptotic pathway, thus making it the most directly involved gene in programmed cell death, whereas upstream genes could influence additional genes or alternative processes. We have clarified this choice in the revised methods section (lines 748-750).

*- The authors mention maintaining (or at least attempting to maintain) a 1:1 sex ratio for the comparative analysis, but it is unclear if this was also done for the *S. araneus* analysis. If not, why? If so, was sex included as a covariate (e.g., a random effect) in the differential expression analysis? Sex-specific expression elevates with group variation and could impact the discovery of differentially expressed genes.*

Regarding the use of sex as a covariate, we acknowledge the concerns raised. In our evolutionary analyses, we maintained a balanced sex ratio within species when possible. EVE models handle the effect of sex on gene expression as intraspecific variation. In shrews, however, we used males exclusively, as females were only found among juvenile individuals. Including those juvenile females would have introduced age effects, with perhaps a larger effect on our results. For the seasonal data, we have now included sex as a covariate in differential expression analyses. However, our design is imbalanced in relation to sex, which we have now discussed in our methods (lines 713-714) and discussion limitations (lines 544-548).

(4) Discussion: The term "adaptive" is used frequently and liberally throughout the discussion. The interpretation of seasonal changes in gene expression as indicators of adaptive evolution should be done cautiously as such changes do not necessarily imply causal or adaptive associations.

Thank you for this insight. We have reviewed our discussion and clarified that adaptations are putative (i.e. lines 146, 285, and 332), and highlighted this in our limitations section.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) I would recommend always spelling out "Dehnel's phenomenon" or even replacing this term (after crediting the DP term) with the more informative "seasonal size plasticity". Every time I saw "DP", I had to remind myself what this referred to. If the authors choose not to do so, please use the acronym consistently (e.g. line 186 has it spelled out).

We have replaced the acronym DP with either the full term or the more informative “seasonal size plasticity” throughout the text.

(2) Line 202: "DEG" has not been defined. Simply add to the line before.

Thank you for this attention to detail. We have added this to the line above (210).

(3) Please add a reference for the "AnAge" tool that was used to determine if samples were pubescent.

Thank you for identifying this oversight. We have now cited the proper paper in line 634.

(4) In the BCL2L1 section in the results, add a callout to Figure 2D.

We have now added a callout to Figure 2D within the results (line 234).

Reviewer #2 (Recommendations for the authors):

(1) Line 122: is associated? These adaptations?

Thank you for identifying that we were missing the words “associated with” here. We have fixed this in the revision.

(2) The first paragraph of the Results should be moved to the methods, except maybe the number of orthologs.

Thank you for this insight. We have removed this portion from the results section.

(3) Why a Bonferroni correction on line 188? That seems too strict.

We agree the Bonferroni correction is strict. Results when using other less strict methods for controlling false discovery rate are also not significant after correction. These corrections can be found within the data, however, we only report on the Bonferroni correction.

(4) Line 427: "is a novel candidate gene for several neurological disorders" needs some references. I see them a couple of sentences later, but that's quite a sentence with no references at the end.

We have added the proper citations for this sentence (line 524).

Reviewer #3 (Recommendations for the authors):

(1) Temporal Expression - Figure 1 and Supplemental Figure 2 and associated text Line176-193:

- The authors report the total number of genes meeting inclusion criteria (>0.5-fold change between any two stages and 2 samples >10 normalized reads), but it would be more informative to also provide the number of genes within each temporal cluster. This would offer a clearer understanding of how gene expression patterns are distributed over time.

Unfortunately, this information is difficult to depict on our figure and would use too much space in the text. We have thus added a description of the range of genes in a new supplemental table depicting this information.

- It is unclear whether quantitative criteria were used to distinguish "developmental shift" clusters from "season shift" clusters. A visual inspection of Supplemental Figure 2 suggests that some clusters (e.g., clusters 2, 8, and to a lesser extent 12) show seasonal variation, not just developmental differences between stages 1 and 2. While clustering helps to visualize expression patterns, it may not be the most appropriate filter in this case, particularly since all "season shift" clusters are later combined in KEGG pathway and GO analyses (Fig. 1B). Using a differential gene expression criterion might be more suitable. For example, do excluded genes show significant log-fold differences between late-stage comparisons?

As previously mentioned, we have now quantified seasonal shifts as large differences in z-score ($\text{abs}(\text{summer juveniles} - \text{summer adults}) > 1.25$) without meaningful interseason variations determined by a second local maximum ($\text{abs}(\text{autumn} - \text{winter}) < 0.5$ and $\text{abs}(\text{winter} - \text{summer}) < 0.5$), and added it to our methods (lines 699-702). We then follow this up with differential expression analyses as described in Figure 2.

- Did the authors perform cluster-specific GO or KEGG pathway enrichment analyses instead of focusing on the combined set of genes across the season shift clusters? While I understand that the small number of genes in each cluster may be limiting, if pathways emerge from cluster-specific analysis, they could provide more detailed insights into the functional significance of these temporal expression patterns. The current analysis picks up relevant pathways for hypothalamic control of homeostasis, which is a useful validation, but this approach might not fully address the study's key hypotheses. Additionally, no corrections for multiple hypothesis testing were applied, as noted in the results. A more refined gene set (e.g., using differential expression criteria, described above) could be more appropriate for these analyses.

We have now included cluster-specific KEGG enrichments as previously described.

(2) Differential expression between shrinkage (stage 2) and regrowth (stage 4) and cell culture targets - Figure 2 and lines 195-227:

- The rationale for selecting BCL2L1 for cell culture experiments should be clarified. While it is part of the apoptosis pathway, several other apoptosis-related genes were identified in the differential gene expression (DGE) analysis, some showing stronger differential expression or shrew-specific branch shifts. Why was BCL2L1 prioritized over these other candidates?

We have now included the reasoning for further validation of BCL2L1 as described above.

- The relevance of the "higher degree" differentially expressed genes needs more explanation. Although this group of genes is highlighted in the results, they are not featured in any subsequent analyses, leaving their importance unclear.

Thank you for this insight. We have removed this from the methods as it is not relevant to subsequent analyses or conclusions.

- The authors mention maintaining (or at least attempting to maintain) a 1:1 sex ratio for the comparative analysis (Line 525), but it is unclear if this was also done for the S. araneus analysis. If so, was sex included as a covariate (e.g., a random effect) in the differential expression analysis?

We have now incorporated information on sex as described above.

(3) Discussion:

The term "adaptive" is used frequently and liberally throughout the discussion, but the authors should be cautious in interpreting seasonal changes in gene expression as indicators of adaptive evolution. Such changes do not necessarily imply causal or adaptive associations, and this distinction should be clearly stated when discussing the results.

Thank you for this feedback and we agree with your conclusion, while a second expression optimum in the shrew lineage is indicative of adaptive expression, we cannot fully determine whether these are caused by genetic or environmental factors, despite careful attention to experimental design. We have highlighted this as a limitation in the discussion.

(4) Minor Editorial Comment:

Line 105: "... maintenance of an energy budgets..." delete "an"

We have removed this grammatical error.

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